

Лабораторная 2

Тема: Настройка протокола STP (IEEE 802.1D)

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"

Полезная информация: схема сохранена на сервере в проекте под именем Menzhulin-lab2-template, можно использовать кнопку Duplicate

Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета

Вначале на каждом PC настроил IP. Так как маршрутизатора нет, то сделал одну сеть - 192.168.1.____ (Далее пример из консоли для одного PC):

```
PC1> ip 192.168.1.10/24
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip

NAME          : PC1[1]
IP/MASK        : 192.168.1.10/24
GATEWAY        : 0.0.0.0
DNS            :
MAC            : 00:50:79:66:68:00
LPORT         : 22000
RHOST:PORT     : 127.0.0.1:22001
MTU            : 1500
```

Итоговые IP:

192.168.1.10/24

192.168.1.20/24

192.168.1.30/24

192.168.1.40/24

192.168.1.50/24

192.168.1.60/24

Далее на каждом коммутаторе включил режим STP

```
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree mode pvst
vIOS-L2-01(config)#end
vIOS-L2-01#
```

На коммутаторе, который я выбрал как корневой, явно установил что это корень:

```
vIOS-L2-01#enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096
vIOS-L2-01(config)#end
vIOS-L2-01#
*Aug  3 02:44:21.237: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
Compressed configuration from 5276 bytes to 2026 bytes[OK]
vIOS-L2-01#
*Aug  3 02:45:00.593: %GRUB-5-CONFIG_WRITING: GRUB configuration is being
updated on disk. Please wait...
*Aug  3 02:45:01.629: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written
to disk successfull
```

```
vIOS-L2-01#show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cf0.fb45.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4097 (priority 4096 sys-id-ext 1)

Address 0cf0.fb45.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr

Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

Тут видно, что Root ID и Bridge ID совпадают, что подтверждает то, что vIOS-L2-01 стал корневым коммутатором.

хоть я и поставил командой приоритет 4096, по факту видим 4097. Это произошло, так как учитывается ещё и номер VLAN 1. $4097 = 4096 + \text{номер VLAN 1}$

`show spanning-tree vlan 1` на всех коммутаторах:

Layer2Switch-2 : show spanning-tree vlan 1

```
vIOS-L2-01>show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol ieee

```

Root ID      Priority      4097
             Address      0cf0.fb45.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

```

```

Bridge ID    Priority      32769 (priority 32768 sys-id-ext 1)
             Address      0cdf.b10e.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	-----

Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Altn	BLK	4	128.5	Shr
Gi1/1	Altn	BLK	4	128.6	Shr
Gi1/2	Altn	BLK	4	128.7	Shr
Gi1/3	Altn	BLK	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

```
vIOS-L2-01>show interfaces status | include Gi2/0
```

```
Gi2/0                                notconnect    1                auto    auto unknown
```

Layer2Switch-3 : show spanning-tree vlan 1

```
vIOS-L2-01>show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    4097
           Address    0cf0.fb45.0000
           Cost        4
           Port        1 (GigabitEthernet0/0)
           Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    0c94.a678.0000
           Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time   300 sec
```

```
Interface                Role Sts Cost          Prio.Nbr Type
-----
```

```
-----
Gi0/0                    Root FWD 4           128.1    Shr
Gi0/1                    Altn BLK 4           128.2    Shr
Gi0/2                    Desg FWD 4           128.3    Shr
Gi0/3                    Desg FWD 4           128.4    Shr
Gi1/0                    Desg FWD 4           128.5    Shr
Gi1/1                    Desg FWD 4           128.6    Shr
```

Layer2Switch-4 : show spanning-tree vlan 1

```
vIOS-L2-01>show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    4097
           Address    0cf0.fb45.0000
           Cost        4
           Port        1 (GigabitEthernet0/0)
           Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    0c1c.8fb8.0000
           Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time   15 sec
```

```
Interface                Role Sts Cost          Prio.Nbr Type
```

```

-----
-----
Gi0/0          Root FWD 4          128.1    Shr
Gi0/1          Altn BLK 4          128.2    Shr
Gi0/2          Desg FWD 4          128.3    Shr
Gi0/3          Desg FWD 4          128.4    Shr
Gi1/0          Desg FWD 4          128.5    Shr
Gi1/1          Desg FWD 4          128.6    Shr

```

Layer2Switch-5 : show spanning-tree vlan 1

```

vIOS-L2-01>show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     0cf0.fb45.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0cb8.0f72.0000
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  15 sec

Interface                Role Sts Cost        Prio.Nbr Type
-----
-----
Gi0/0          Root FWD 4          128.1    Shr
Gi0/1          Altn BLK 4          128.2    Shr
Gi0/2          Desg FWD 4          128.3    Shr
Gi0/3          Desg FWD 4          128.4    Shr
Gi1/0          Desg FWD 4          128.5    Shr
Gi1/1          Desg FWD 4          128.6    Shr

```

На каждом

Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать

PC1 ping other

```

PC1> ping 192.168.1.20

84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=2.310 ms

```

```
84 bytes from 192.168.1.20 icmp_seq=2 ttl=64 time=6.659 ms
84 bytes from 192.168.1.20 icmp_seq=3 ttl=64 time=9.891 ms
84 bytes from 192.168.1.20 icmp_seq=4 ttl=64 time=13.700 ms
84 bytes from 192.168.1.20 icmp_seq=5 ttl=64 time=4.304 ms
```

```
PC1> ping 192.168.1.30
```

```
84 bytes from 192.168.1.30 icmp_seq=1 ttl=64 time=18.280 ms
84 bytes from 192.168.1.30 icmp_seq=2 ttl=64 time=29.972 ms
84 bytes from 192.168.1.30 icmp_seq=3 ttl=64 time=21.115 ms
84 bytes from 192.168.1.30 icmp_seq=4 ttl=64 time=20.817 ms
84 bytes from 192.168.1.30 icmp_seq=5 ttl=64 time=20.122 ms
```

```
PC1> ping 192.168.1.40
```

```
84 bytes from 192.168.1.40 icmp_seq=1 ttl=64 time=20.932 ms
84 bytes from 192.168.1.40 icmp_seq=2 ttl=64 time=12.345 ms
84 bytes from 192.168.1.40 icmp_seq=3 ttl=64 time=16.089 ms
84 bytes from 192.168.1.40 icmp_seq=4 ttl=64 time=20.875 ms
84 bytes from 192.168.1.40 icmp_seq=5 ttl=64 time=23.614 ms
```

```
PC1> ping 192.168.1.50
```

```
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=12.727 ms
84 bytes from 192.168.1.50 icmp_seq=2 ttl=64 time=11.636 ms
84 bytes from 192.168.1.50 icmp_seq=3 ttl=64 time=17.007 ms
84 bytes from 192.168.1.50 icmp_seq=4 ttl=64 time=7.499 ms
84 bytes from 192.168.1.50 icmp_seq=5 ttl=64 time=22.557 ms
```

```
PC1> ping 192.168.1.60
```

```
84 bytes from 192.168.1.60 icmp_seq=1 ttl=64 time=13.255 ms
84 bytes from 192.168.1.60 icmp_seq=2 ttl=64 time=26.776 ms
84 bytes from 192.168.1.60 icmp_seq=3 ttl=64 time=17.759 ms
84 bytes from 192.168.1.60 icmp_seq=4 ttl=64 time=23.047 ms
84 bytes from 192.168.1.60 icmp_seq=5 ttl=64 time=20.313 ms
```

PC2 ping other (except PC 1)

```
Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.
```

```
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
```

```
Press '?' to get help.
```

```
Executing the startup file
```

```
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0
```

```
PC2> ping 192.168.1.30
```

```
84 bytes from 192.168.1.30 icmp_seq=1 ttl=64 time=12.499 ms
84 bytes from 192.168.1.30 icmp_seq=2 ttl=64 time=20.372 ms
84 bytes from 192.168.1.30 icmp_seq=3 ttl=64 time=22.640 ms
84 bytes from 192.168.1.30 icmp_seq=4 ttl=64 time=33.613 ms
84 bytes from 192.168.1.30 icmp_seq=5 ttl=64 time=25.632 ms
```

```
PC2> ping 192.168.1.40
```

```
PC2> ping 192.168.1.30
```

```
84 bytes from 192.168.1.30 icmp_seq=1 ttl=64 time=12.499 ms
84 bytes from 192.168.1.30 icmp_seq=2 ttl=64 time=20.372 ms
84 bytes from 192.168.1.30 icmp_seq=3 ttl=64 time=22.640 ms
84 bytes from 192.168.1.30 icmp_seq=4 ttl=64 time=33.613 ms
84 bytes from 192.168.1.30 icmp_seq=5 ttl=64 time=25.632 ms
```

```
PC2> ping 192.168.1.40
```

```
84 bytes from 192.168.1.40 icmp_seq=1 ttl=64 time=20.401 ms
84 bytes from 192.168.1.40 icmp_seq=2 ttl=64 time=24.757 ms
84 bytes from 192.168.1.40 icmp_seq=3 ttl=64 time=18.891 ms
84 bytes from 192.168.1.40 icmp_seq=4 ttl=64 time=27.573 ms
84 bytes from 192.168.1.40 icmp_seq=5 ttl=64 time=14.435 ms
```

```
PC2> ping 192.168.1.50
```

```
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=25.155 ms
84 bytes from 192.168.1.50 icmp_seq=2 ttl=64 time=19.653 ms
84 bytes from 192.168.1.50 icmp_seq=3 ttl=64 time=24.769 ms
84 bytes from 192.168.1.50 icmp_seq=4 ttl=64 time=37.675 ms
84 bytes from 192.168.1.50 icmp_seq=5 ttl=64 time=15.337 ms
```

```
PC2> ping 192.168.1.60
```

```
84 bytes from 192.168.1.60 icmp_seq=1 ttl=64 time=16.781 ms
84 bytes from 192.168.1.60 icmp_seq=2 ttl=64 time=13.911 ms
84 bytes from 192.168.1.60 icmp_seq=3 ttl=64 time=22.458 ms
84 bytes from 192.168.1.60 icmp_seq=4 ttl=64 time=10.137 ms
84 bytes from 192.168.1.60 icmp_seq=5 ttl=64 time=14.210 ms
```

PC3 ping other (except PC1, PC2)

```
PC3> ping 192.168.1.40
```

```
84 bytes from 192.168.1.40 icmp_seq=1 ttl=64 time=4.130 ms
84 bytes from 192.168.1.40 icmp_seq=2 ttl=64 time=2.137 ms
84 bytes from 192.168.1.40 icmp_seq=3 ttl=64 time=4.707 ms
84 bytes from 192.168.1.40 icmp_seq=4 ttl=64 time=2.128 ms
84 bytes from 192.168.1.40 icmp_seq=5 ttl=64 time=2.323 ms
```

```
PC3> ping 192.168.1.50
```

```
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=8.374 ms
84 bytes from 192.168.1.50 icmp_seq=2 ttl=64 time=13.127 ms
84 bytes from 192.168.1.50 icmp_seq=3 ttl=64 time=15.662 ms
84 bytes from 192.168.1.50 icmp_seq=4 ttl=64 time=13.664 ms
84 bytes from 192.168.1.50 icmp_seq=5 ttl=64 time=21.193 ms
```

```
PC3> ping 192.168.1.60
```

```
84 bytes from 192.168.1.60 icmp_seq=1 ttl=64 time=21.699 ms
84 bytes from 192.168.1.60 icmp_seq=2 ttl=64 time=21.605 ms
84 bytes from 192.168.1.60 icmp_seq=3 ttl=64 time=5.378 ms
84 bytes from 192.168.1.60 icmp_seq=4 ttl=64 time=21.993 ms
84 bytes from 192.168.1.60 icmp_seq=5 ttl=64 time=12.221 ms
```

PC4 ping other (except PC1, PC2, PC3)

```
PC4> ping 192.168.1.50
```

```
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=25.750 ms
84 bytes from 192.168.1.50 icmp_seq=2 ttl=64 time=12.467 ms
84 bytes from 192.168.1.50 icmp_seq=3 ttl=64 time=12.792 ms
84 bytes from 192.168.1.50 icmp_seq=4 ttl=64 time=4.303 ms
84 bytes from 192.168.1.50 icmp_seq=5 ttl=64 time=17.180 ms
```

```
PC4> ping 192.168.1.60
```

```
84 bytes from 192.168.1.60 icmp_seq=1 ttl=64 time=9.012 ms
84 bytes from 192.168.1.60 icmp_seq=2 ttl=64 time=8.174 ms
84 bytes from 192.168.1.60 icmp_seq=3 ttl=64 time=25.490 ms
```



```
84 bytes from 192.168.1.60 icmp_seq=4 ttl=64 time=17.686 ms
84 bytes from 192.168.1.60 icmp_seq=5 ttl=64 time=7.328 ms
```

PC5 ping PC6

```
PC5> ping 192.168.1.60
```

```
84 bytes from 192.168.1.60 icmp_seq=1 ttl=64 time=17.839 ms
84 bytes from 192.168.1.60 icmp_seq=2 ttl=64 time=2.582 ms
84 bytes from 192.168.1.60 icmp_seq=3 ttl=64 time=1.909 ms
84 bytes from 192.168.1.60 icmp_seq=4 ttl=64 time=1.842 ms
84 bytes from 192.168.1.60 icmp_seq=5 ttl=64 time=1.306 ms
```

(пункт 3) На изображении схемы отметить VID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

“eltex_education_2026/module4/labs/lab2_chema.png” не может быть найдена.

При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет

The image shows a Wireshark capture of STP Hello BPDUs. The packet list shows multiple BPDUs from different sources. The packet details for frame 13 show a Spanning Tree Protocol Configuration BDU with Root ID 4096, Root Path Cost 0, and Bridge ID 4096.

No.	Time	Source	Destination	Protocol	Length	Info
13	13.316646	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
16	15.563285	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
22	16.912109	0c:df:b1:0e:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/1/0c:b8:0f:72:00:00 Cost = 4 Port = 0x8002
23	17.466695	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
42	21.714384	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
43	21.715756	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
44	21.717029	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
45	22.002134	0c:df:b1:0e:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 32768/1/0c:df:b1:0e:00:00 Cost = 0 Port = 0x8002
47	22.539800	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
48	22.770353	0c:df:b1:0e:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:df:b1:0e:00:00 Cost = 0 Port = 0x8002
49	22.772804	0c:df:b1:0e:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:df:b1:0e:00:00 Cost = 0 Port = 0x8002
50	22.774642	0c:df:b1:0e:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:df:b1:0e:00:00 Cost = 0 Port = 0x8002
53	23.591039	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
54	24.187458	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
55	24.188858	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
56	24.190305	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
58	24.885406	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
59	25.462394	0c:df:b1:0e:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:df:b1:0e:00:00 Cost = 0 Port = 0x8002

Frame 13: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Port identifier: 0x8002

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

-> В захвате видно что в списке поначалу появляются разные Root ID. Как я понял, в момент начала захвата, коммутаторы обменивались BDPUs и пересчитывали роли после включения портов (подобная картина может ещё также быть после изменения

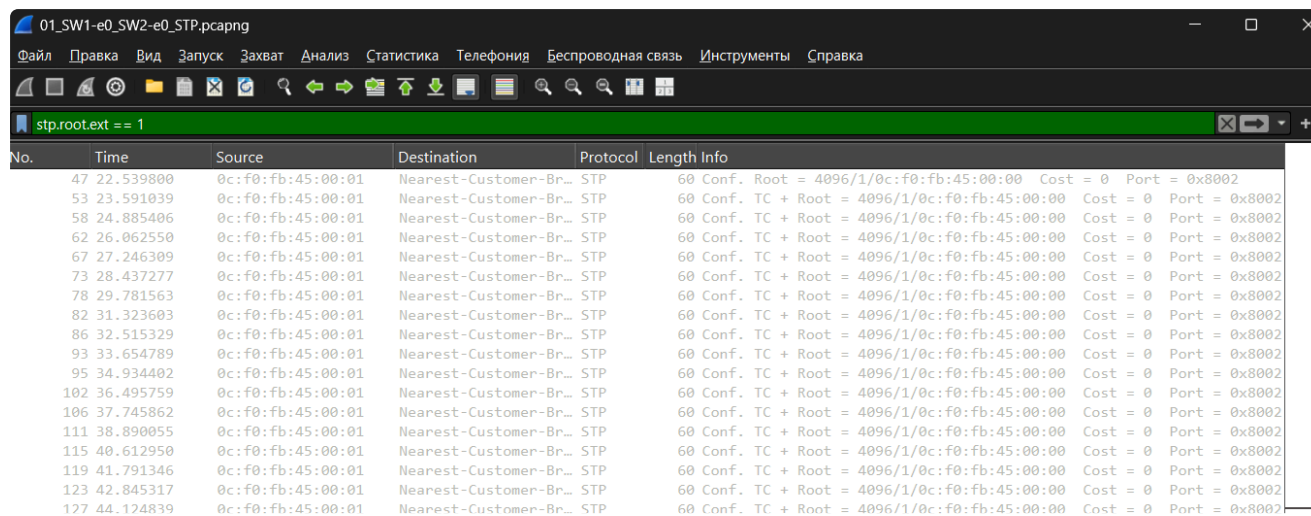
конфигурации).

-> видно что wireshark показывает несколько VLAN

```
Root = 32768/100/...  
Root = 32768/200/...  
Root = 32768/300/...
```

Это из-за того, что PVST поддерживает отдельное дерево для каждой VLAN. Нам интересно сейчас только VLAN 1, поэтому поставлен доп фильтр:

```
stp.root.ext == 1
```



No.	Time	Source	Destination	Protocol	Length	Info
47	22.539800	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
53	23.591039	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
58	24.885406	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
62	26.062550	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
67	27.246309	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
73	28.437277	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
78	29.781563	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
82	31.323603	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
86	32.515329	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
93	33.654789	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
95	34.934402	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
102	36.495759	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
106	37.745862	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
111	38.890055	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
115	40.612950	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
119	41.791346	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
123	42.845317	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002
127	44.124839	0c:f0:fb:45:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8002

-> Тут уже видно что кадры отправляются уже настоящим (определённым мной) корнем:

```
Source: 0c:f0:fb:45:00:01  
Root: 4096/1/0c:f0:fb:45:00:00  
Cost: 0
```

у корня стоимость пути до самого себя всегда равна нулю.

```
Spanning Tree Protocol
  Protocol Identifier: Spanning Tree Protocol (0x0000)
  Protocol Version Identifier: Spanning Tree (0)
  BPDU Type: Configuration (0x00)
  BPDU flags: 0x01, Topology Change
    0... .... = Topology Change Acknowledgment: No
    .... ...1 = Topology Change: Yes
  Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00
    Root Bridge Priority: 4096
    Root Bridge System ID Extension: 1
    Root Bridge System ID: 0c:f0:fb:45:00:00 (0c:f0:fb:45:00:00)
  Root Path Cost: 0
  Bridge Identifier: 4096 / 1 / 0c:f0:fb:45:00:00
    Bridge Priority: 4096
    Bridge System ID Extension: 1
    Bridge System ID: 0c:f0:fb:45:00:00 (0c:f0:fb:45:00:00)
  Port identifier: 0x8002
  Message Age: 0
  Max Age: 20
  Hello Time: 2
  Forward Delay: 15
```

-> Hello time равно 2. Из захвата тоже видно, что интервал примерно ≈2с.

-> Далее таблица линков:

Линия	Отправитель BPDU	Bridge ID отправителя	Root Path Cost	Интервал
Sw1 e0 <-> Sw2 e0	Switch1	4097/0cf0.fb45.0000	0	≈2 с
Sw1 e1 <-> Sw2 e1	Switch1	4097/0cf0.fb45.0000	0	≈2 с
Sw1 e2 <-> Sw3 e0	Switch1	4097/0cf0.fb45.0000	0	≈2 с
Sw1 e3 <-> Sw3 e1	Switch1	4097/0cf0.fb45.0000	0	≈2 с
Sw1 e4 <-> Sw4 e0	Switch1	4097/0cf0.fb45.0000	0	≈2 с
Sw1 e5 <-> Sw4 e1	Switch1	4097/0cf0.fb45.0000	0	≈2 с
Sw1 e6 <-> Sw5 e0	Switch1	4097/0cf0.fb45.0000	0	≈2 с
Sw1 e7 <-> Sw5 e1	Switch1	4097/0cf0.fb45.0000	0	≈2 с
Sw2 e2 <-> Sw3 e2	Switch3	32768/0c94.a678.0000	4	≈2 с

Линия	Отправитель BPDU	Bridge ID отправителя	Root Path Cost	Интервал
Sw2 e3 <-> Sw3 e3	Switch3	32768/0c94.a678.0000	4	≈2 с
Sw3 e4 <-> PC1 e0	Switch3	32768/0c94.a678.0000	4	≈2 с
Sw3 e5 <-> PC2 e0	Switch3	32768/0c94.a678.0000	4	≈2 с
Sw2 e4 <-> Sw4 e2	Switch4	32768/0c1c.8fb8.0000	4	≈2 с
Sw2 e5 <-> Sw4 e3	Switch4	32768/0c1c.8fb8.0000	4	≈2 с
Sw2 e6 <-> Sw5 e2	Switch5	32768/0cb8.0f72.0000	4	≈2 с
Sw2 e7 <-> Sw5 e3	Switch5	32768/0cb8.0f72.0000	4	≈2 с
Sw4 e4 <-> PC3 e0	Switch4	32768/0c1c.8fb8.0000	4	≈2 с
Sw4 e5 <-> PC4 e0	Switch4	32768/0c1c.8fb8.0000	4	≈2 с
Sw5 e4 <-> PC5 e0	Switch5	32768/0cb8.0f72.0000	4	≈2 с
Sw5 e5 <-> PC6 e0	Switch5	32768/0cb8.0f72.0000	4	≈2 с

-> В некоторых случаях, отправитель BPDU также Layer2Switch-1. Это из-за того, что заблокированный порт Switch2 (например) принимает BPDU, но не пропускает обычный трафик.

SW1 e1 DP/FWD → SW2 e1 ALT/BLK

04_SW1-e3_SW3-e1_STP.pcapng [Layer2Switch-1 Ethernet3 to Layer2Switch-3 Ethernet1]

ФайлПравкаВидЗапускЗахватАнализСтатистикаТелефонияБеспроводная связьИнструментыСправка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
16	3.305830	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
20	5.141400	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
24	6.291714	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
28	7.475165	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
33	8.625363	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
38	9.745298	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
41	10.896018	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
49	12.060595	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
52	14.087712	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
57	15.240193	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
61	16.375822	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
66	17.527856	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
71	18.684549	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
75	19.850541	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
80	21.014605	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
84	22.181908	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
89	23.348345	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004
92	24.488753	0c:f0:fb:45:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8004

Frame 38: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Port identifier: 0x8004

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

0000 01 80 c2 00 00 00 0c f0 fb 45 00

0010 03 00 00 00 00 00 10 01 0c f0 fb

0020 00 00 10 01 0c f0 fb 45 00 00 80

0030 02 00 0f 00 00 00 00 00 00 00 80

05_SW1-e4_SW4-e0_STP.pcapng [Layer2Switch-1 Ethernet4 to Layer2Switch-4 Ethernet0]

ФайлПравкаВидЗапускЗахватАнализСтатистикаТелефонияБеспроводная связьИнструментыСправка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
5	1.168351	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
9	2.333444	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
13	3.470987	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
18	4.611598	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
22	5.714539	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
26	6.856232	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
31	8.023074	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
35	9.153861	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
40	10.303675	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
44	11.448221	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
51	12.625520	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
55	14.653419	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
61	15.834287	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
67	17.007295	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
69	18.271114	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
75	19.412630	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005
78	20.571378	0c:f0:fb:45:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8005

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Port identifier: 0x8005

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

0000 01 80 c2 00 00 00 0c f0 fb 45 00

0010 03 00 00 00 00 00 10 01 0c f0 fb

0020 00 00 10 01 0c f0 fb 45 00 00 80

0030 02 00 0f 00 00 00 00 00 00 00 80

06_SW1-e5_SW4-e1_STP.pcapng [Layer2Switch-1 Ethernet5 to Layer2Switch-4 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
5	0.653355	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
8	1.794427	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
16	2.942862	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
20	4.953669	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
25	6.093425	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
30	7.270291	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
33	8.437279	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
39	9.558063	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
42	10.735267	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
48	11.951619	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
50	13.196480	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
56	14.322169	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
59	15.485017	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
66	16.689315	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
68	17.882999	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
74	19.079499	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
78	20.302488	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006
85	21.541673	0c:f0:fb:45:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8006

Frame 48: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Port identifier: 0x8006

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

07_SW1-e6_SW5-e0_STP.pcapng [Layer2Switch-1 Ethernet6 to Layer2Switch-5 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
94	23.869790	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
98	25.014371	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
102	26.153056	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
106	27.325598	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
114	28.459526	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
118	30.577594	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
122	31.807316	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
128	33.008808	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
130	34.126314	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
136	35.266316	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
139	36.422714	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
145	37.597156	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
147	38.820623	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
155	40.038427	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
157	41.198182	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
163	42.337725	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007
165	43.529872	0c:f0:fb:45:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8007

Frame 122: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Port identifier: 0x8007

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

08_SW1-e7_SW5-e1_STP.pcapng [Layer2Switch-1 Ethernet7 to Layer2Switch-5 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
2	0.919135	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
8	2.083850	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
10	3.227643	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
16	4.443964	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
19	5.629668	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
25	6.782653	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
31	7.947136	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
36	9.375829	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
41	10.557719	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
45	11.758594	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
49	12.899158	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
53	14.041310	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
57	15.203362	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
62	16.441125	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
66	17.585954	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
70	18.724435	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
74	19.896593	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008
82	21.030983	0c:f0:fb:45:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 0 Port = 0x8008

Frame 31: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

- Protocol Identifier: Spanning Tree Protocol (0x0000)
- Protocol Version Identifier: Spanning Tree (0)
- BPDU Type: Configuration (0x00)
- BPDU flags: 0x00
- Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00
- Root Path Cost: 0
- Bridge Identifier: 4096 / 1 / 0c:f0:fb:45:00:00
- Port identifier: 0x8008
- Message Age: 0
- Max Age: 20
- Hello Time: 2
- Forward Delay: 15

09_SW2-e2_SW3-e2_STP.pcapng [Layer2Switch-3 Ethernet2 to Layer2Switch-2 Ethernet2]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
5	0.583835	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
8	1.700710	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
13	2.790826	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
17	3.911198	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
22	5.046469	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
25	6.144893	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
27	7.254661	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
35	8.345559	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
37	9.457950	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
43	10.616540	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
46	11.678771	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
52	12.794269	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
54	13.936022	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
60	15.097779	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
62	16.207312	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
68	17.310058	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
70	18.481159	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
77	19.603785	0c:94:a6:78:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003

Frame 22: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

- Protocol Identifier: Spanning Tree Protocol (0x0000)
- Protocol Version Identifier: Spanning Tree (0)
- BPDU Type: Configuration (0x00)
- BPDU flags: 0x00
- Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00
- Root Path Cost: 4
- Bridge Identifier: 32768 / 1 / 0c:94:a6:78:00:00
- Bridge Priority: 32768
- Bridge System ID Extension: 1
- Bridge System ID: 0c:94:a6:78:00:00 (0c:94:a6:78:00:00)
- Port identifier: 0x8003
- Message Age: 2
- Max Age: 20
- Hello Time: 2
- Forward Delay: 15

10_SW2-e3_SW3-e3_STP.pcapng [Layer2Switch-3 Ethernet3 to Layer2Switch-2 Ethernet3]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
3	1.111877	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
10	2.201521	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
12	3.314164	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
18	4.473494	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
21	5.535251	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
27	6.649261	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
29	7.789693	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
35	8.951266	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
37	10.061384	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
43	11.164126	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
46	12.335463	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
53	13.457905	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
55	14.586180	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
60	15.733491	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
63	16.887743	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
67	18.031223	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
72	19.166619	0c:94:a6:78:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004

Frame 18: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:94:a6:78:00:00

Port identifier: 0x8004

Message Age: 1

Max Age: 20

Hello Time: 2

Forward Delay: 15

11_SW3-e4_PC1-e0_STP.pcapng [Layer2Switch-3 Ethernet4 to PC1 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
3	0.643676	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
4	1.839608	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
5	3.079212	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
6	4.317277	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
7	5.520978	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
8	6.692998	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
9	7.903065	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
12	9.097426	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
14	10.371961	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
15	11.619917	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
16	13.500035	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
17	14.702876	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
18	15.890858	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
19	17.018874	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
20	18.223013	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
21	19.493444	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
22	20.680839	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
24	21.860286	0c:94:a6:78:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005

Frame 3: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:94:a6:78:00:00

Port identifier: 0x8005

Message Age: 1

Max Age: 20

Hello Time: 2

Forward Delay: 15

Снимок экрана скопирован в
Автоматически сохраняются
экрана.

Разметка и общ

14_SW2-e5_SW4-e3_STP.pcapng [Layer2Switch-2 Ethernet5 to Layer2Switch-4 Ethernet3]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
6	1.085337	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
9	2.199167	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
15	3.304052	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
20	4.398232	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
22	5.545345	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
28	6.646743	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
30	7.724146	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
36	8.833012	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
38	9.924349	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
44	11.021442	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
46	12.131776	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
53	13.244525	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
56	14.326583	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
61	15.429071	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
65	16.528845	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
68	17.629185	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
73	18.702033	0c:1c:8f:b8:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004

Frame 28: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:1c:8f:b8:00:00

Port identifier: 0x8004

Message Age: 2

Max Age: 20

Hello Time: 2

Forward Delay: 15

15_SW2-e6_SW5-e2_STP.pcapng [Layer2Switch-2 Ethernet6 to Layer2Switch-5 Ethernet2]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
9	1.124903	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
10	2.269703	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
17	3.355904	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
18	4.469392	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
25	5.569469	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
26	6.703182	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
34	7.876875	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
35	8.989794	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
42	10.125434	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
44	11.249317	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
50	12.342869	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
55	13.439705	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
59	14.550825	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
63	15.661946	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
67	16.801202	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
73	17.912952	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003
78	19.020604	0c:b8:0f:72:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8003

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:b8:0f:72:00:00

Port identifier: 0x8003

Message Age: 1

Max Age: 20

Hello Time: 2

Forward Delay: 15

16_SW2-e7_SW5-e3_STP.pcapng [Layer2Switch-2 Ethernet7 to Layer2Switch-5 Ethernet3]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
8	1.099835	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
9	2.234063	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
17	3.406332	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
18	4.523429	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
25	5.654252	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
27	6.778469	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
33	7.871809	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
37	8.967677	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
41	10.080542	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
46	11.195614	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
50	12.329116	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
55	13.443009	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
60	14.551844	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
64	15.686760	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
68	16.798400	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
72	17.907761	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004
76	19.033130	0c:b8:0f:72:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8004

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:b8:0f:72:00:00

Port identifier: 0x8004

Message Age: 1

Max Age: 20

Hello Time: 2

Forward Delay: 15

17_SW4-e4_PC3-e0_STP.pcapng [Layer2Switch-4 Ethernet4 to PC3 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
2	0.358854	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
3	1.574022	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
4	2.765974	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
5	3.955385	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
6	5.195427	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
7	6.493743	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
8	7.713937	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
9	8.903014	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
10	10.085738	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
13	11.369342	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
14	12.553288	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
15	13.729963	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
16	14.909700	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
17	16.078341	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
18	17.806573	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
19	19.029714	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
20	20.214238	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
21	21.443896	0c:1c:8f:b8:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005

Frame 2: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:1c:8f:b8:00:00

Port identifier: 0x8005

Message Age: 1

Max Age: 20

Hello Time: 2

Forward Delay: 15

18_SW4-e5_PC4-e0_STP.pcapng [Layer2Switch-4 Ethernet5 to PC4 Ethernet0]

ФайлПравкаВидЗапускЗахватАнализСтатистикаТелефонияБеспроводная связьИнструментыСправка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
2	1.239961	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
3	2.538396	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
4	3.758227	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
5	4.948121	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
6	6.130493	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
8	7.414044	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
9	8.598208	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
10	9.774845	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
11	10.954310	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
12	12.122953	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
13	13.851734	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
14	15.074480	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
15	16.258895	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
16	17.489199	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
18	18.702421	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
19	19.916336	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006
20	21.087735	0c:1c:8f:b8:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8006

Frame 5: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:1c:8f:b8:00:00

Port identifier: 0x8006

Message Age: 1

Max Age: 20

Hello Time: 2

Forward Delay: 15

000001 80 c2 00 00 00 0c 1c 8f b8 00

001003 00 00 00 00 00 10 01 0c f0 fb

002000 04 80 01 0c 1c 8f b8 00 00 80

003002 00 0f 00 00 00 00 00 00 00 00

19_SW5-e4_PC5-e0_STP.pcapng [Layer2Switch-5 Ethernet4 to PC5 Ethernet0]

ФайлПравкаВидЗапускЗахватАнализСтатистикаТелефонияБеспроводная связьИнструментыСправка

stp.root.ext == 1

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
2	1.176736	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
3	2.436217	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
4	3.687301	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
5	4.816120	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
6	6.026374	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
7	7.197722	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
8	8.389311	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
10	9.615650	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
11	11.617536	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
12	12.840063	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
13	14.086621	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
14	15.270176	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
15	16.489585	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
16	17.635288	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
17	18.859199	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
18	20.009666	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005
23	21.221840	0c:b8:0f:72:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:f0:fb:45:00:00 Cost = 4 Port = 0x8005

Frame 6: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f0:fb:45:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:b8:0f:72:00:00

Port identifier: 0x8005

Message Age: 1

Max Age: 20

Hello Time: 2

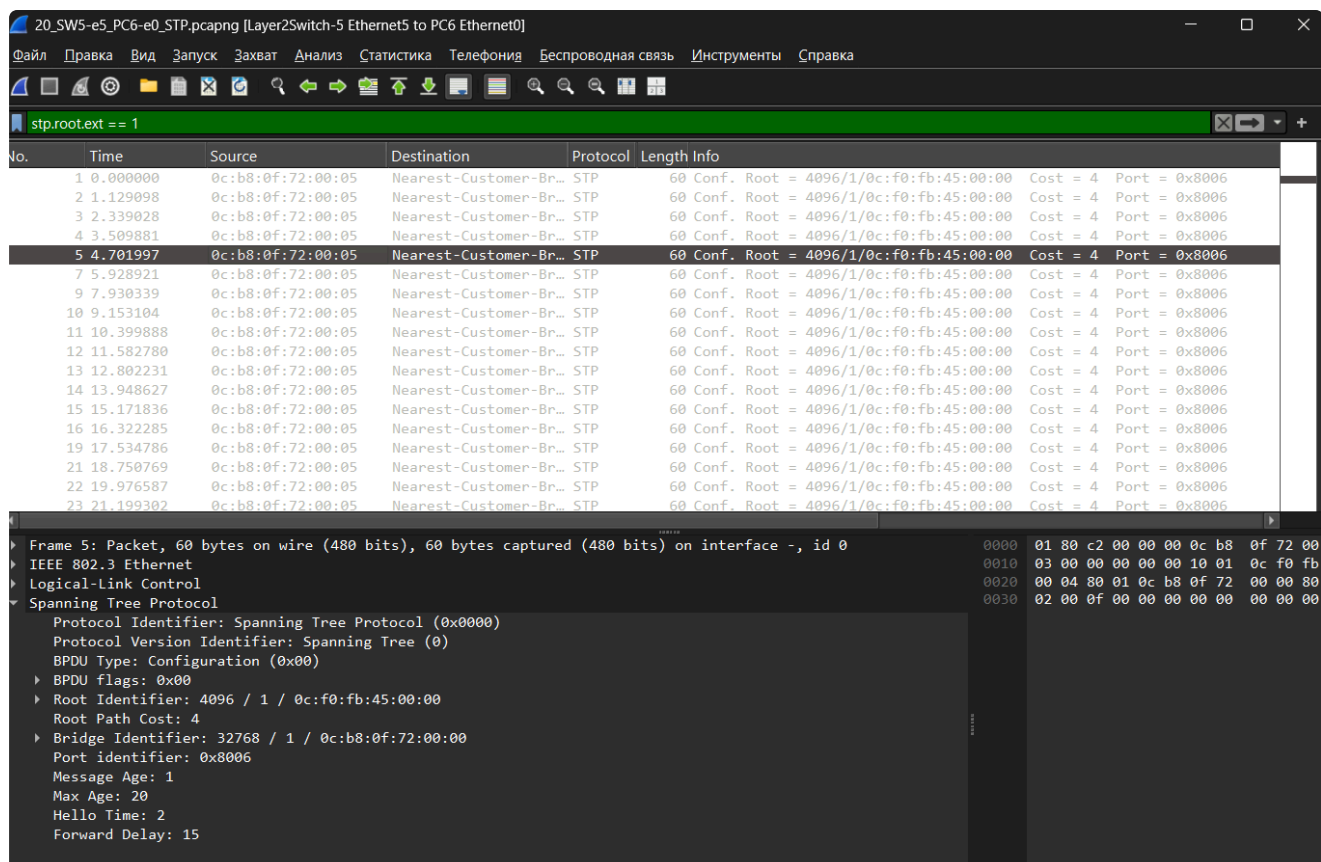
Forward Delay: 15

000001 80 c2 00 00 00 0c b8 0f 72 00

001003 00 00 00 00 00 10 01 0c f0 fb

002000 04 80 01 0c b8 0f 72 00 00 80

003002 00 0f 00 00 00 00 00 00 00 00



Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл

(пункт 3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

-> Я выбрал Switch3 и изменил стоимость для GigabitEthernet0/0:

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
    Root ID    Priority    4097
              Address    0cf0.fb45.0000
              Cost        4
              Port        1 (GigabitEthernet0/0)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
              Address    0c94.a678.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   15 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	-----

Gi0/0	Root	LRN	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Desg	LRN	4	128.3	Shr
Gi0/3	Desg	LRN	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

```

vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#interface GigabitEthernet0/0
vIOS-L2-01(config-if)#spanning-tree vlan 1 cost 20
vIOS-L2-01(config-if)#end

vIOS-L2-01#copy running-config startup-config

```

-> После изменения стоимости:

```

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     0cf0.fb45.0000
             Cost        4
             Port        2 (GigabitEthernet0/1)
             Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec

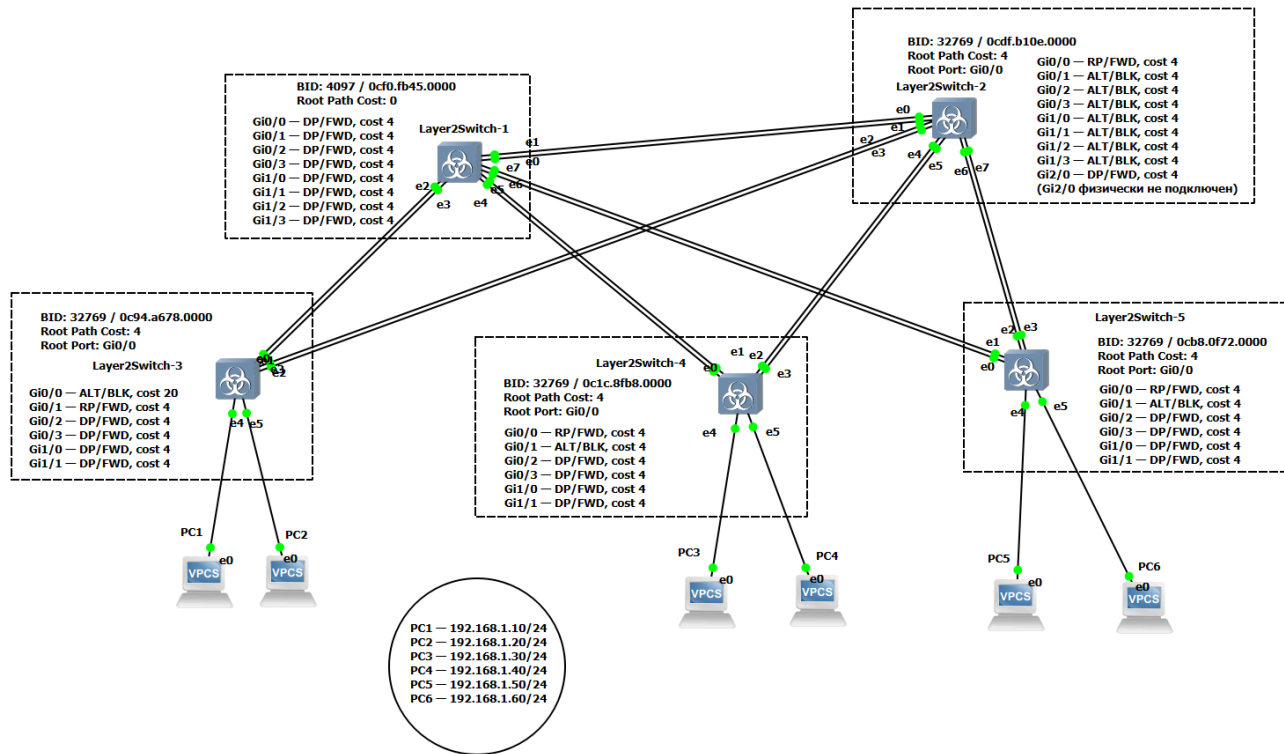
  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0c94.a678.0000
             Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
             Aging Time   300 sec

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	-----

Gi0/0	Altn	BLK	20	128.1	Shr
Gi0/1	Root	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

-> изменённая схема:



Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

```
Layer2Switch-1.txt  PC1.txt
Файл  Изменить  Просмотр
no ipv6 cef
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 1 priority 4096
!
vlan internal allocation policy ascending
!
vlan 100
  name VLAN100
!
vlan 200,300
!
```

Для коммутаторов:

```
terminal length 0
show running-config
```

Для PC:


```
show ip
```

Конфигурации сохранены:

```
lab2_config/  
├─ Layer2Switch-1.txt  
├─ Layer2Switch-2.txt  
├─ Layer2Switch-3.txt  
├─ Layer2Switch-4.txt  
├─ Layer2Switch-5.txt  
├─ PC1.txt  
├─ PC2.txt  
├─ PC3.txt  
├─ PC4.txt  
├─ PC5.txt  
└─ PC6.txt
```

***Оptionальное задание:** заменить STP на RSTP (IEEE 802.1w), повторить 1-6, отметить резервные порты в п.3 и п.5, отличие работы протокола RSTP от протокола STP в п.4